

## BCNF (Boyce Codd NF).

Explanation ①  $\alpha \rightarrow \beta$   
(Prime) (Non-Prime) X Not allowed in 2NF.

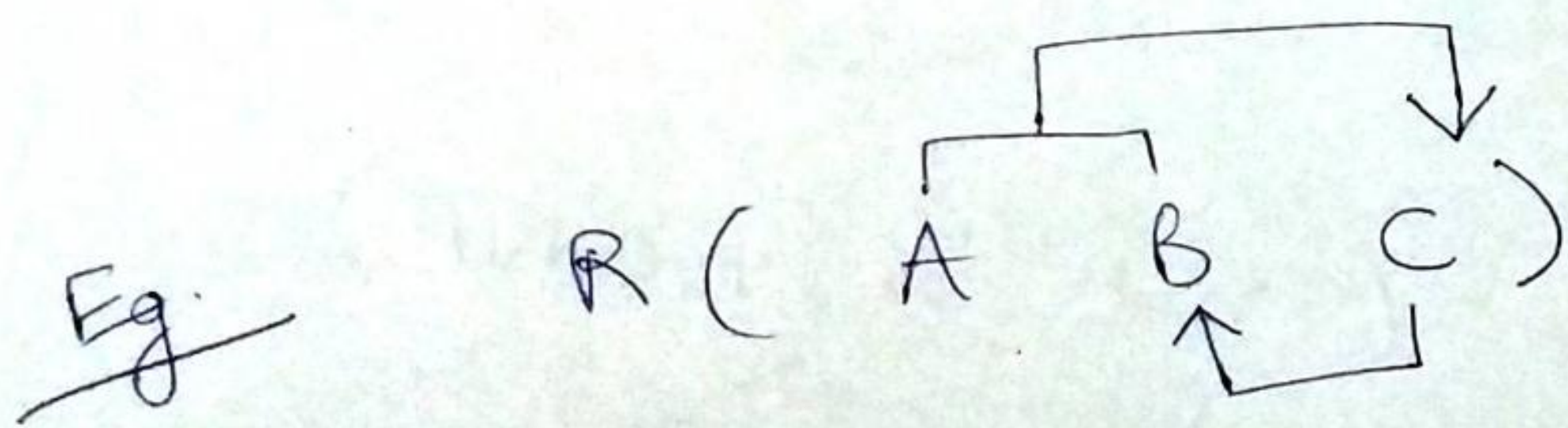
②  $\alpha \rightarrow \beta$   
(Non-Prime) (Non-Prime) X Not allowed in 3NF, bcz this is Transitive nature

③  $\alpha \rightarrow \beta$   
(Prime / Non Prime) (Prime) → This is not present in 2NF as well as in 3NF. So BCNF handles these cases.

Defn It is an advance version of 3NF that is why it is referred as 3.5NF.

BCNF is stricter than 3NF.

A table complies with BCNF if it is in 3NF and for every functional dependency  $X \rightarrow Y$ ,  $X$  should be super key of the table.



$F = \{$   
 $AB \rightarrow C$   
 $C \rightarrow B$   
 $\}$

$(A)^+ \times$

$(AB)^+ = ABC \checkmark$

$(AC)^+ = ACB \checkmark$

Step 1 check whether this relation is in 2NF or not

$AB \rightarrow C$  (Not a partial dependency)

$C \rightarrow B$  (Not a partial dependency)

(Prime) (Prime)

P.T.O

Step 2: check 3NF.

$AB \rightarrow C$  (Not Transitive  
bcz  $NP \rightarrow NP$   
is not there)

$C \rightarrow B$  (Not transitive)

(Prime) (Prime)

~~Defn.~~

Step 3:  $AB \rightarrow C$  (This is in BCNF)

↙  
Super key

$C \rightarrow B$  (not in BCNF)

(not super  
key)

↘ this can be P/NP.

Since Super key and candidate key  
are entitled to derive every other  
attribute.

# Decomposition

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A relation 'R' can be decomposed into a collection of relation schemas to eliminate some of the anomalies in original relation R.

## ① Lossless Join Decomposition

Let R is a relation and has set of F.D's 'F' over R. The decomposition of R into  $R_1$  &  $R_2$  is lossless w.r.t F if  $R_1 \bowtie R_2 = R$ . (In other words, if all attributes of relation are covered in the decomposition & covered in the decomposition & there should be some common attribute in the decomposed relation which must be the candidate key of either of the relation.)

↓  
(natural join)

② Lossy Decomposition → extra tuples.

there should be some common attribute in the decomposed relation which must be the candidate key of either of the relation.

NOTE

if  $(R_1 \bowtie R_2)$  forms a super key in either  $R_1$  or  $R_2$ , then it is lossless.

## Solved Examples on Decomposition

Q1:  $R(ABC)$  decomposed in  $R_1(AB)$  and  $R_2(BC)$ . Check whether lossless or lossy.

Ans:  $A \rightarrow B$  (is given)

$R_1 \cap R_2 = B$  ( $\because B$  is not a superkey in either  $R_1$  or  $R_2$ ).  
So, decomposition is Lossy.

Q2:  $R(ABC)$ ,  $F: \{A \rightarrow B\}$  decomposition of  $R_1(AB)$ ,  $R_2(AC)$  is lossless or lossy.

Ans:  $R_1 \cap R_2 = A$  ( $A$  is a key in relation  $R_1$ ).

So, decomposition is lossless.